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AUTOMATED CALIBRATION OF THE METRIC-LANDSAT EVAPOTRANSPIRATION PROCESS[1]

Richard G. Allen, Boyd Burnett, William Kramber, Justin Huntington, Jeppe Kjaersgaard, Ayse Kilic, Carlos Kelly, and Ricardo Trezza[2]

ABSTRACT: A remaining challenge to applying satellite-based energy-balance algorithms for operational estimation of evapotranspiration (ET) is the calibration of the energy-balance model. Customized calibration for each image date is generally required to overcome biases associated with radiometric accuracy of the image, uncertainties in aerodynamic features of the landscape, background thermal conditions, and model assumptions. The *CIMEC* process (calibration using inverse modeling at extreme conditions) is an endpoint calibration procedure where near extreme conditions in the image are identified where the ET can be estimated and assigned. In the Mapping EvapoTranspiration at high Resolution with Internalized Calibration (METRIC™) energy-balance model, two endpoints represent the dry and wet ends of the ET spectrum. Generally, user-intervention is required to select locations in the image to produce best accuracy. To bring the METRIC and similar processes into the domain of less experienced operators, a consistent, reproducible, and dependable statistics-based procedure is introduced where relationships between vegetation amount and surface temperature are used to identify a subpopulation of locations (pixels) in an image that may best represent the calibration endpoints. This article describes the background and logic for the statistical approach, how the statistics were developed, area of interest requirements and assumptions, adjustment for dry conditions in desert climates, and implementation in a common image processing environment (ERDAS Imagine).

(KEY TERMS: evapotranspiration; energy balance; remote sensing; calibration.)

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INTRODUCTION

Application of remote sensing algorithms solving the energy balance using high resolution satellite imagery has proven useful for establishing estimates of actual evapotranspiration, ET_a, and relative evapotranspiration for large populations of fields and water users (Bastiaanssen *et al.*, 1998; Tasumi *et al.*, 2005a, b; Allen *et al.*, 2007a, b). The use of a surface

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[2]Respectively, Professor of Water Resources Engineering (Allen), University of Idaho, Kimberly Research and Extension Center, 3793 N 3600 E, Kimberly, Idaho 83341; Engineer (Burnett), PPL Montana, LLC, Butte, Montana 59701; Remote Sensing Analyst (Kramber), Idaho Department of Water Resources, Boise, Idaho 83720; Assistant Research Professor (Huntington), Desert Research Institute, Reno, Nevada 89512; Assistant Professor (Kjaersgaard), Department of Agricultural Engineering, South Dakota State University, Brookings, South Dakota 57007; Associate Professor of Civil Engineering (Kilic), School of Natural Resources, University of Nebraska-Lincoln, Lincoln, Nebraska 68583-0973; and Remote Sensing Technician (Kelly) and Research Associate Professor (Trezza), University of Idaho, Kimberly Research and Extension Center, Kimberly, Idaho 83341 (E-Mail/Allen: rallen@kimberly.uidaho.edu).

energy balance to determine ET has strong advantages, primarily in that specific vegetation type does not need to be known, and the energy balance can detect reduced ET caused by water shortage, salinity, or frost that may not correlate with vegetation amount. In addition, thermally based energy-balance methods can detect evaporation from bare soil. A number of operational processing models use a *CIMEC* process (calibration using inverse modeling at extreme conditions) (Bastiaanssen *et al.*, 1998; Allen *et al.*, 2011a, b) that identifies extreme or near extreme conditions in the image to use as endpoints, and where the ET can be independently estimated and assigned. In the Mapping EvapoTranspiration at high Resolution with Internalized Calibration (METRIC™) energy-balance procedure the two endpoints represent the dry and wet ends of the ET spectrum. Generally, user-intervention and selection is required to select locations in the image representing these conditions and to assign ET estimates (Allen *et al.*, 2007b, 2011a, b).

The *CIMEC* process is generally required to produce image-specific calibration of the energy-balance process to overcome biases associated with radiometric accuracy of the satellite image, uncertainties in aerodynamic features of the landscape and associated wind speed, sun angle, background thermal conditions, and assumptions made in the model. Some sort of surface-air temperature difference or near surface air temperature difference is required in surface energy-balance models to estimate sensible heat flux, which in turn, is used to estimate ET as a residual of the energy balance. However, differences in processes for determining air temperature (generally based on measurements at a weather station) and surface temperature (T_s) (determined radiantly from the satellite) can create considerable biases between these two estimates. In addition, air temperature measurements recorded within several meters of the ground contain strong feedbacks to the energy partitioning (i.e., amount of ET) at the underlying surface, which are variable among weather stations. Inverse modeling and the employment of a near surface temperature gradient *vs.* T_s relationship in METRIC is necessary to overcome these and other important biases.

The *CIMEC* process employs population dynamics associated with vegetation amount and T_s because of the difficulty in tying the surface temperature endpoints to any commonly available ground-based measurements such as air temperature. As described in the next section, a near surface temperature difference, dT, is used in place of a T_s-air temperature difference, to drive the determination of sensible heat flux. The calibration of the dT *vs.* T_s function is determined within the *CIMEC* process. In employing the *CIMEC* process in METRIC, the assumption is made that confidence and accuracy are increased when associating the maximum ET (and the corresponding dT) for full-cover vegetation conditions with the alfalfa reference ET, as opposed to associating maximum ET with estimates of potential energy availability. Alfalfa reference ET can be determined with the ASCE standardized Penman-Monteith method (ASCE-EWRI, 2005) with relatively good accuracy. In addition, at the dry endpoint, a continuous soil water balance using precipitation as an input provides good accuracy and dependability, in the *CIMEC* process, for estimating residual ET for bare soil conditions.

Human intervention in the *CIMEC* process generally produces best accuracy, especially if done by an experienced, thoughtful operator. However, selecting appropriate and representative anchor pixels in an image can be a time-consuming and generally iterative process and may be an intimidating task for less experienced users. As a consequence, the endpoint pixel selection process may exhibit some operator dependency or human error that is then incorporated into the final map of ET. An automated selection of the anchor pixels may provide for more consistent calibration that can reduce the user dependency of the calibration, especially for less experienced users, and may improve general estimation accuracy and processing speed, provided the automation is consistently accurate.

The approach developed and reported here utilizes tendencies and relationships between vegetation amount, represented by a vegetation index, the surface albedo, and T_s to statistically identify a subpopulation of locations (pixels) in an image that best represents the calibration endpoints. The automated method is intended to function as a first-cut approach to calibrating a Landsat image to produce spatial ET maps. The result may be a suitable stopping point for some users and applications. More experienced users working on applications where the accuracy needs to be higher than that produced by the automated selection method may elect to refine the selections by the automated method. One objective of this article is to describe the setup of the proposed method and show performance results of the method. An application using ERDAS Imagine Modelmaker software is described, however, the method can be readily implemented into other scripting languages such as Python (Morton *et al.*, 2013). The calibration procedure isolates a subpopulation of candidate anchor pixels from which the user can select the pixels to apply. In the ERDAS application, a color coding directs the user toward more preferred pixels. Results from applications in Montana, Idaho, and elsewhere indicate that the method performs relatively well in identifying anchor pixels having characteristics and resulting calibrations that compare relatively well to those obtained using pixels selected independently by experienced users.

BACKGROUND ON METRIC

The METRIC model is a satellite image processing model that estimates ET as the residual of the energy balance (Allen *et al.*, 2002, 2007b; Tasumi *et al.*, 2005b):

$$\mathrm{LE} = R_\mathrm{n} - G - H \tag{1}$$

where LE is latent heat flux density, R_n is net radiation, G is soil heat flux density, and H is sensible heat flux density. The METRIC model is designed as an operational, engineering approach to retrieval of ET images and was developed primarily for application with Landsat imagery to achieve high ET product resolution (30 m) that is useful for monitoring water consumption at field (human) scales. The 30 m resolution ET is necessary to produce aggregated ET for individual fields and for relatively narrow riparian systems along streams. 30 m resolution ET is produced following sharpening of the thermal pixels using a vegetation index (Allen *et al.*, 2011a). Thermal pixels are 120 m for Landsats 4 and 5, 60 m for Landsat 7, and 100 m for Landsat 8. The thermal resolution is coarser than that for reflected energy because fewer photons are available at the satellite in the thermal range so that a larger source area is required to sustain a workable signal-to-noise ratio. Calibration of METRIC is generally applied to a complete 160 km wide Landsat scene (Allen *et al.*, 2007a, b). In METRIC, R_n is calculated by solving the radiation balance as described by Allen *et al.* (2007b). Soil heat flux is the rate of heat conducted into soil and vegetation, and is estimated in METRIC (Allen *et al.*, 2011a) from H, R_n, T_s and the Normalized Difference Vegetation Index, NDVI.

Sensible heat flux is the convective heat loss from the surface to the air created by a near surface temperature gradient. H is estimated in METRIC using a one-dimensional, blended aerodynamic, temperature gradient-based method:

$$H = \frac{\rho\, c_p\, \mathrm{d}T}{r_\mathrm{ah}} \tag{2}$$

where ρ is air density (kg/m^3), c_p is air specific heat, (J/kg/K), dT (K) is the temperature difference between two heights (z_1 and z_2) in a near surface blended layer, and r_ah is the aerodynamic resistance to heat transport (s/m) between z_1 and z_2. The use of near surface dT rather than estimates for air and surface temperature, directly, in Equation (2) is done to reduce impacts of bias in temperature, as explained earlier.

CALIBRATION OF METRIC

The *CIMEC* procedure used in METRIC (Allen *et al.*, 2011a) is traceable to Bastiaanssen (1995) and Bastiaanssen *et al.* (1998) and is used to calibrate the dT function of Equation (2). The *CIMEC* process is applied by inverting Equation (2) for dT after solving Equation (1) for H for the two extreme endpoint conditions in the image representing nearly full LE and nearly zero LE. In METRIC, LE for the full LE condition is estimated using alfalfa-based reference ET, $\mathrm{ET_r}$, represented by the American Society of Civil Engineers (ASCE) standardized Penman-Monteith equation (ASCE-EWRI, 2005). The $\mathrm{ET_r}$ is calculated using quality controlled hourly weather data (Allen, 1996; ASCE-EWRI, 2005) from a weather station residing in the image area and collecting data from a relatively well-watered environment so that ground-atmosphere feedback conditions expected by the ASCE reference method are satisfied. The use of alfalfa reference $\mathrm{ET_r}$ provides a consistent and relatively accurate basis for calibration of the "high" endpoint of an image for full vegetation cover, which is a relatively readily identifiable condition within the image.

In areas where data from a "well-watered" weather station are not available, the user may have to use data from weather stations that may exhibit dryness effects associated with lack of soil water. The higher temperature and lower humidity measurements recorded at dry sites will increase the estimate for $\mathrm{ET_r}$. The false increases in $\mathrm{ET_r}$ will cause overestimation in the time-integrated ET determined by integrating ET images over time, for example over monthly and growing season periods, and should be corrected using some type of conditioning method or other empirical method, for example as recommended by ASCE-EWRI (2005) (see Allen *et al.*, 2007b). Bias in the value used for $\mathrm{ET_r}$, however, is not critical to the calibration and application of METRIC during the instantaneous energy balance existing at the time of the satellite overpass. Nor is the process sensitive to bias in the value used for wind speed during the METRIC calibration. This is because it is the fraction of $\mathrm{ET_r}$, termed $\mathrm{ET_rF}$, that is "mandated" during the *CIMEC* process, and as shown later, $\mathrm{ET_rF}$ adheres to user mandated values. These mandated values are then reproduced by the calibrated METRIC model, regardless of the

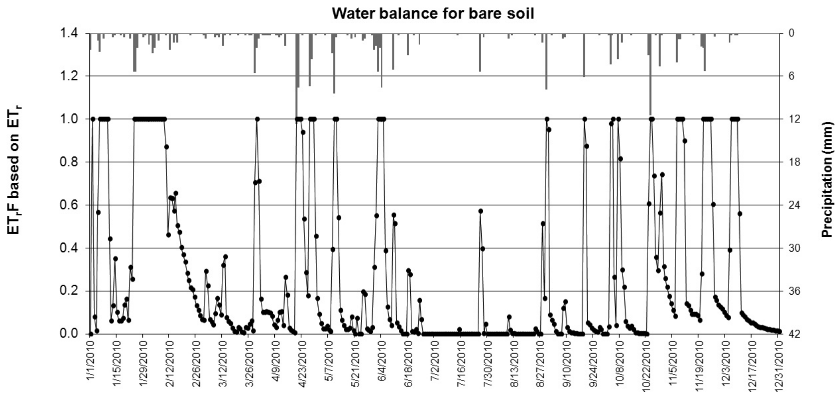

FIGURE 1. Daily Relative Evaporation from Bare Soil, Expressed as the Fraction of Reference ET, ET_rF, for the Christmas Valley, Oregon Area During 2010.

magnitude of the ET_r or wind speed values. The only impact to ET_rF by ET_r occurs in the area about midway between the two extremes of ET_rF, where differences in buoyancy correction can occur when the magnitude of ET_r (and LE) or wind speed change.

LE for the low endpoint condition is estimated in METRIC using a daily or hourly soil water balance and evaporation model that accounts for any residual evaporation from bare soil stemming from antecedent precipitation events. Any daily or hourly process model can be utilized to estimate the residual evaporation. In many METRIC applications (Allen *et al.*, 2011a), the relatively simple "slab" model of FAO-56 (Allen *et al.*, 1998) is used, with the skin-layer evaporation extension of Allen (2011). Typical simulation results from the soil water balance model are shown in Figure 1. The estimation of LE for the bare soil condition is assigned to an identified bare soil condition in the processed image. The bare soil condition is preferably located in an agricultural region where soil conditions are likely to be similar to water holding and tilth characteristics assumed in the evaporation model and where soil heat flux characteristics are similar to those assumed by the G algorithms that are typically developed for tilled agricultural soils. The value for LE from the process model is placed into the inverses of Equations (1) and (2) in METRIC to solve for dT for the bare soil condition.

The dT for each pixel in the Landsat scene is estimated as an indexed function of radiometric surface temperature from the satellite, T_s, where a simple linear relationship is assumed to exist between dT and T_s, based on the two extreme conditions within the scene:

$$\mathrm{d}T = aT_{s_{\mathrm{DEM}}} + b \tag{3}$$

where a and b are calibration coefficients. In METRIC, T_s is delapsed to $T_{s_{\mathrm{DEM}}}$ via a common arbitrary elevation datum using an image-specific lapse rate, generally between 6 and 10 K/km (Allen *et al.*, 2007b, 2011a). The delapsing adjusts for change in T_s caused by warming or cooling of the air mass with elevation so that any remaining variation in $T_{s_{\mathrm{DEM}}}$ can be associated with variation in evaporation and the surface energy balance. The linearity of the dT *vs*. $T_{s_{\mathrm{DEM}}}$ function is a major assumption in METRIC. Research by Bastiaanssen (1995), comparisons of final ET estimates over a range of dT values (Tasumi *et al.*, 2005b; Allen *et al.*, 2007a) and theoretical arguments (Allen *et al.*, 2007b) suggest that this assumption can apply to a wide range of conditions.

In METRIC, agricultural areas are generally targeted when identifying the extreme endpoint conditions because of their importance in water consumption and management, because vegetation in intensively managed agricultural systems is more uniform at the 30 m scale compared to natural systems, and because aerodynamic roughness, soil heat flux, and emissivity algorithms inside METRIC are more likely to have the highest accuracy under these conditions. For example, desert soils tend to run warmer than the same soil under bare, agricultural conditions due to differences in soil tilth, structure, crusting, layering, dryness, and sealing. The calibration endpoints ideally represent the conditions in agricultural fields having full and actively transpiring vegetation or having a bare or very low vegetation cover with known residual evaporation from the soil. The anchor pixel representing full vegetation cover has the characteristics of green vegetation and its T_s and dT are low due to evaporative cooling. This condition is commonly referred to as the "cold" condition. The anchor pixel representing a dry, bare endpoint has the characteristics of a bare, tilled soil and its dT and T_s are high due to radiometric heating of the surface. This condition is commonly referred to as the "hot" condition.

The use of alfalfa-based ET_r in the estimation of LE at the cold anchor pixel is generally effective and accurate in representing a near upper limit on ET in the

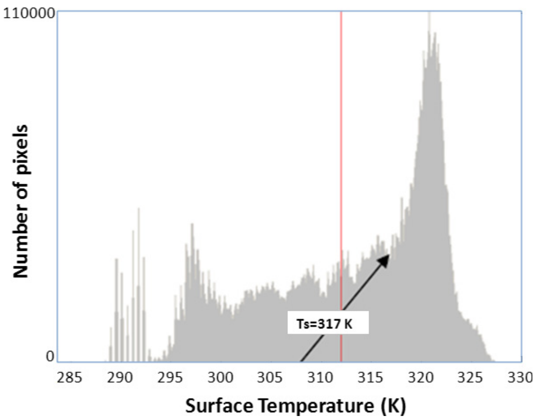

FIGURE 2. Land Surface Temperature Histogram for June 4, 2000 for Path 39, Row 30 of a Landsat Image (created in ERDAS IMAGINE).

energy-balance calibration. The use of ET_r calculated by the ASCE standardized Penman-Monteith method is especially useful and necessary in arid and semiarid climates having regional advection of heat energy and dry air. Regional advection can increase LE to twice the available energy ($R_n - G$) (Allen *et al.*, 2011a) and must be accounted for. An advantage of using alfalfa ET_r is that its approximation of a near upper limit on ET, constrained by energy availability, makes the review of a produced ET image straightforward in that the fraction of reference ET, ET_rF, tends to converge at 1.0, on average, for full cover, well-watered vegetation. ET_rF is synonymous with the commonly used alfalfa-based crop coefficients.

The calibration of METRIC and the statistical identification of calibration endpoints leverages the wide range of land surface temperature (LST) exhibited by vegetation and soil and caused by the amount of available energy ($R_n - G$) and amount of latent heat flux, LE, that impacts LST via the conversion of available energy to LE and the associated evaporative cooling. A typical frequency distribution of $T_{s_{DEM}}$ is shown in Figure 2 from an early June Landsat image in southeastern Idaho. High LST is associated with low LE. The large peak in the histogram from approximately 317 K to 330 K represents the large number of desert pixels within the image having low LE and high temperature. The bare, dry agricultural pixels in the early June image averaged 314-317 K (arrow in the figure). These temperatures are several K lower than those for desert due to effects of tillage of agricultural soils on crust reduction and increased G and due to increased LE from residual soil moisture. The user defined temperature for the hot pixel condition was determined to be 315.2 K during manual calibration of METRIC, based on iterative review of values and distribution of ET_rF for bare or nearly bare soil conditions across the image, particularly in agricultural areas.

AUTOMATED CALIBRATION METHOD

The statistical procedure to select the cold and hot anchor pixels was developed for METRIC in 2004 (Allen, internal UI report) and refined by Burnett (2007), Kjaersgaard and Allen (2009), and Allen (2012, internal UI report). A summary of the statistical procedure is given in Table 1. Similar procedures have been tested by Irmak *et al.* (2011) and Morton *et al.* (2013). Basically, the cold pixel candidate has been found to typically lie in the coolest, most vegetated 1% of areas within the image dominated by agricultural fields (comprising the coldest 20% of the highest 5% vegetated pixels) and to have an ET_rF value of 1.05 when fully vegetated. The hot pixel candidate has been found to generally lie in the hottest, least vegetated 2% of the image (the warmest 20% of the lowest 10% vegetated pixels) and to have an ET_rF value associated with the ET_rF for a bare soil condition, adjusted for any residual vegetation and evaporation. The T_s associated with the hot pixel is additionally adjusted in dry climates if precipitation during the previous several months has been very low so that the upper soil profile is very dry. This adjustment is described in a following section. Depending on the area of application, it may be necessary to identify and delineate larger agricultural areas to avoid dominance by, for example, desert pixels in the anchor pixel selection process.

The reasons for assigning $ET_rF = 1.05$ to the cold pixel of METRIC were listed by Allen *et al.* (2007b). In essence, the value represents a scaling of the variance of the population of ET_rF about the mean value of 1.0 for full-cover agricultural crops, and is consistent with the value used during manual calibration where the user's focus in an image typically gravitates toward the cooler end of T_s for full-cover conditions. The 20 and 5% statistics were developed by observing manual METRIC calibrations in Idaho and reproducing the T_s selection that was associated with the condition where $ET_rF = 1.05$. As noted, the product of 20 and 5% in general directs the isolation process to the coolest 1% of pixels that, due to the nature of the area of interest (AOI), described below, are also the more fully vegetated fraction of the AOI. Experience in calibration tends to support that about 1% of a total population in a predominately well-watered agricultural area will have an ET_rF value that exceeds 1.05. The exceedance is caused by natural variation in vegetation wetness, stomatal conductance, and albedo and inherent error variance in satellite image values and in model estimates.

Figures 3 left and 3 right illustrate the relationship between ET_rF and two common vegetation

TABLE 1. Summary of the Statistical Procedure for Automated Selection of Cold and Hot Anchor Pixels.

Anchor Pixel	Step No.	Procedure
AOI	1a	Outline an Area of Interest (AOI) or several AOIs in the image that contain primarily agricultural land-use classes (in semiarid climates, these should be irrigated agriculture). A target of >80% agricultural pixels is desired, with some inclusions of roadways, homesteads, small towns, nonagricultural areas, etc. permitted. It is desirable to locate the AOI(s) within 20-30 km of the weather station from which the ET_r is determined for calibration
	1b	Alternatively, select all pixels or a subset (grid) of all pixels within 20-30 km of the weather station that have an agricultural (or irrigated agricultural [in dry climates]) land use (LU) (for example, LU = 81, 82 for the National Land Cover Database (NLCD)). In option 1b, filtering may be desired to ensure that selected pixels are away from field boundaries (see the text). In the case of the hot pixel selection, a grass rangeland (in arid climates) LU, for example LU 71 in the NLCD system, can be included to ensure sufficient number of pixels having "bare soil" conditions during mid growing season
	2	Follow the described approach in the text for filtering pixels to eliminate effects of field edges on values for thermal pixels and relationships between T_s and NDVI. The selection of 1a or 1b AOI and edge filtering will impact the statistics and results
Cold	1	Identify the top 5% NDVI pixels within the predefined AOI that reside within 20-30 km of the weather station for which ET_r is calculated
	2	From the group isolated in step 1, identify the coldest 20% pixels, regarding the delapsed surface temperature, $T_{s_{DEM}}$. $T_{s_{DEM}}$ is used to remove elevation-related artifacts in T_s. Calculate the average $T_{s_{DEM}}$ among the coldest 20% $T_{s_{DEM}}$ (The average NDVI of this 1% of the population is used in step 6 as $NDVI_1$)
	3	Recommended cold pixel candidates are those pixels that have a $T_{s_{DEM}}$ within ± 0.2 K of the average $T_{s_{DEM}}$ and albedo within ± 0.02 of Equation (7). The 0.02 window of the albedo can be relaxed if a larger number of candidate pixels is desired
	4	Apply the post processing filter for albedo using Equation (7) to isolate candidates that are most similar to conditions assumed for the alfalfa ET_r reference crop
	5	One pixel is selected from the group in 3, based on the homogeneity among its neighboring pixels and distance from the weather station. The user should visually review the location to ensure that it is among a relatively uniformly vegetated group of pixels, away from the center of a center pivot irrigated field, and among pixels having similar thermal conditions. The image attributes for the selected pixel are used in the calibration of the dT
	6	Assign ET_rF to the cold candidate in proportion to the vegetation amount (necessary for periods of low NDVI, otherwise, $ET_rF_{cold} = 1.05$ is recommended): $$ET_rF_{cold} = 1.54NDVI_1 - 0.10 \text{ for } NDVI_1 < 0.75 \quad (4a)$$ $$ET_rF_{cold} = 1.05 \text{ otherwise} \quad (4b)$$ $NDVI_1$ is from step 2
Hot	1	Identify the lowest 10% NDVI pixels within the predefined AOI (for the AOI 1b approach, one can include a nonagricultural LU such as grassland to increase the number of bare or nearly bare soil pixels (for example, if the final pool from step 3 has fewer than 50 pixels))
	2	From the group isolated in step 1 identify the hottest 20% $T_{s_{DEM}}$. Calculate the average $T_{s_{DEM}}$ among the hottest 20% $T_{s_{DEM}}$
	3	Adjust the average temperature from step 2 using Equation (8)
	4	One pixel is selected from the group in 2 and 3 that has a $T_{s_{DEM}} \pm 0.2$ K of the average, adjusted $T_{s_{DEM}}$. The user should visually review the location to ensure that it is among a relatively uniformly vegetated group of pixels and within 10-20 km of the precipitation gauge used to estimate the ET_rF_{bare} from the water balance model. The closer the hot pixel location is to the precipitation gauge, the more likely the history of precipitation is similar to that recorded at the gauge and the more likely that the estimated ET_rF_{bare} is representative for the pixel location
	5	Assign the ET_rF_{hot} based on the ET_rF_{bare}: $$ET_rF_{hot} = (f_c ET_rF_{cold} + (1 - f_c) ET_rF_{bare}) \quad (5)$$ where $$f_c = \left(\frac{NDVI_{hot} - NDVI_{bare}}{NDVI_{full} - NDVI_{bare}}\right) \quad (6)$$ where $NDVI_{hot}$ is the NDVI of the hot pixel that has been selected. $NDVI_{bare}$ is the NDVI representing bare soil for the image area and $NDVI_{full}$ is about 0.8 for full vegetation, but can be adjusted to fit the image. Equations (5) and (6) are applied only if it is likely that there are few truly "bare" pixels in the AOI, for example, if the AOI happens to be rangeland or if it contains agriculture, but is during July when most fields are vegetated
	6	Use all properties of the selected pixel in the calibration of the dT

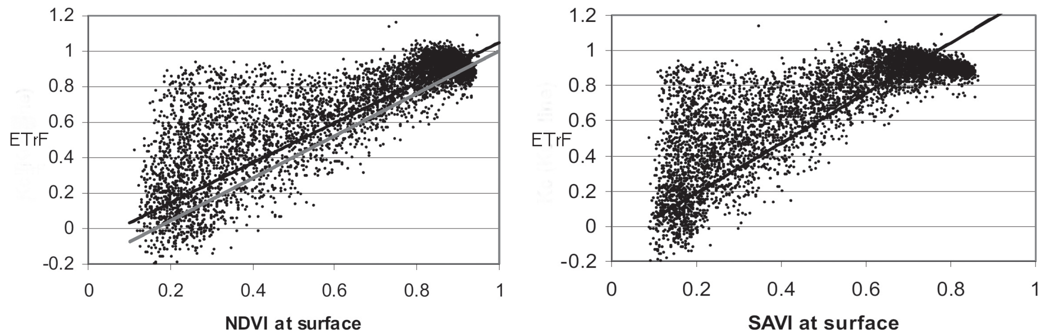

FIGURE 3. ET_rF Estimated by METRIC for Several Hundred Potato Fields in Southern Idaho over a 100-Day Period During Year 2000 *vs.* NDVI (left) and *vs.* SAVI (right), Where NDVI and SAVI Were Calculated Using Surface Reflectance and the Lines in the Figures Represent ET_rF for Dry Soil Surface (basal) Conditions. In left figure, the black line is a general crop line: basal ETrF = 1.13 NDVI − 0.08 and the grey line is a potato-specific line: basal ETrF = 1.19 NDVI − 0.19. (After Burnett, 2007).

indices, NDVI and soil adjusted vegetation index, SAVI. These figures show ET_rF estimated by METRIC for several hundred potato fields in southern Idaho over a 100-day period, where maximum ET_rF for potatoes tended to average about 0.9, which is 0.1 lower than for other southern Idaho crops, consistent with lysimeter-based findings by Wright (1982). Figure 3 left shows the distribution of near maximum ET_rF over a range of near maximum NDVI, both having variances of about 0.05-0.1 for ET_rF and about 0.05 for NDVI. The NDVI was computed using surface reflectance (Tasumi *et al.*, 2005a, 2008) so that values are somewhat greater than if "top of atmosphere" reflectances had been used. The variances centered on the average maximum values support the 5% statistics used to specify the ET_rF = 1.05 and the upper 5% NDVI for the nearly full-cover condition associated with the cold pixel endpoint. A point of interest in Figure 3 right is the wider spread of SAVI under near full-cover conditions when ET_rF is at maximum. The wider spread is caused by the tendency for the SAVI calculation to remain "unsaturated" over a wider, high range of leaf area, whereas, the NDVI tends to "saturate" at lower values (typically 3) for leaf area index. Because the ET_rF value, which is essentially the same as the crop coefficient, K_c, tends to also saturate, or reach a maximum value, at about the same amount of leaf area as does NDVI, NDVI tends to be a better, more linear means for specifying ET_rF compared to SAVI. The "general K_{cb}" and "K_{cb} custom" lines in the left figure represent "basal" relationships between ET_rF and NDVI when the soil surface is expected to be dry so that most of the ET signal is from transpiration. The "general" curve was derived for application to all primary agricultural crops in south-central Idaho and the "custom" curve was determined specifically for potatoes.

The lines in Figures 3 left and 3 right represent ET_rF for dry soil surface (basal) conditions. Data points, representing ET_rF for individual fields, lying well above the basal lines reflect the additional evaporation from wet soil. Negative values for ET_rF in Figure 3 demonstrate some of the uncertainty associated with determination of very low, and even zero values for ET from surface energy balance, where small or zero values for ET are calculated from the differences among relatively large values for R_n, H, and G. These particular pixels likely had ET near zero, and the distribution of negative ET_rF from zero provides an indication of the relative uncertainty in all estimates of ET using METRIC, especially for low ET. In this case, the standard deviation of negative ET_rF from zero was about 0.05.

Table 1 recommends a downward adjustment to the ET_rF = 1.05 value during periods of universally low vegetation cover, for example during winter periods in high latitudes. The adjustment is made based on the mean NDVI of the population of the 1% of pixels that are isolated during the first two steps of the cold pixel selection. The relationship was derived from samples of ET_rF and NDVI over a number of processed Landsat images in paths 39 and 40 in southern Idaho and represents relationships common to row-crop and forage crop agriculture.

As described previously, a daily "slab" evaporation model (Allen *et al.*, 1998; Allen, 2011) is typically used to estimate the residual ET_rF from bare soil that stems from prior precipitation events. A nonzero ET_rF may be common to agricultural soils due to upward flow of liquid or vapor from deeper, moist soil, supplied by previous irrigation or wet off-season periods. For example, Wright (1982) and FAO-56 (Allen *et al.*, 1998) generally established a lower limit on ET_rF for bare soil (ET_rF_{bare}) of 0.1 to account for residual, diffusive evaporation. Therefore, during calibration of METRIC, a value for ET_rF for the hot pixel condition is often set to 0.05 or 0.10 (Allen *et al.*, 2007b, 2011a). However, when long periods occur without significant precipitation during periods with high evaporative demand, bare soil evaporation will approach zero asymptotically and the value for ET_rF_{bare} is set to 0.0. The transformation of ET_rF_{bare} into the ET_rF to assign to the hot condition, ET_rF_{hot}, is given in Table 1 via Equations (5) and (6). The transformation is made using the fraction of surface covered by vegetation, estimated from NDVI. The transformation is necessary in situations where no or very few areas of bare soil are available in an image, for example during midseason in intensively cultivated regions.

The values for NDVI representing bare and full-cover conditions vary with soil characteristics, with values of 0.15 and 0.8 being common. The value of NDVI for the lower 2% that results from steps 1 and 2 of the hot pixel process, $NDVI_2$, is the mean for those isolated pixels.

It is apparent from histograms of LST such as shown in Figure 2 that any statistical-based method for anchor pixel selection must include measures to avoid the inclusion of large areas of desert or other nonagricultural pixels that will vary from scene to scene. Use of a consistent type of AOI helps to ensure that nonagricultural inclusions comprise a consistent percentage of the AOI.

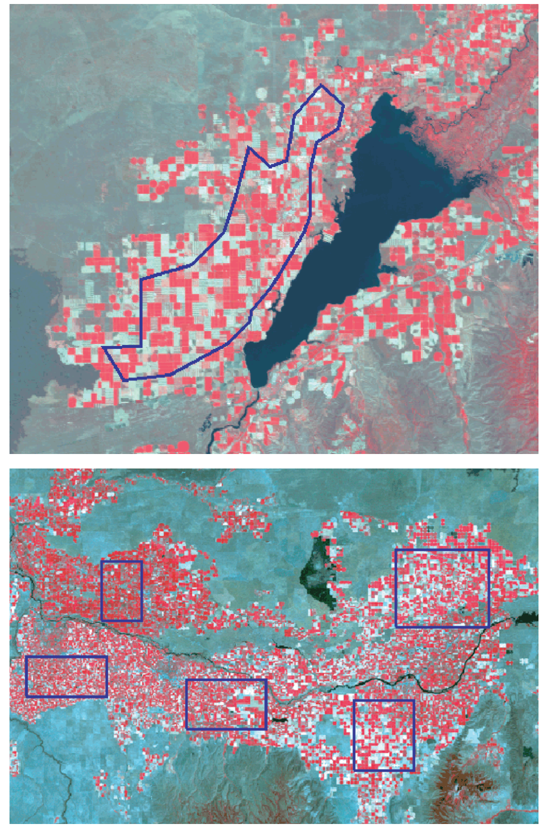

FIGURE 4. Examples of Areas of Interest (blue outlines) in Southeastern Idaho (top) and South-Central Idaho (bottom) Delineating Approximately 150 km^2 Areas Used for Isolating Cold and Hot Pixel Candidates. Locations of weather stations used during METRIC calibration are shown as yellow dots.

Area of Interest

The first step in the automated selection process is to identify an AOI from which to select the endpoint candidates. Table 1 specifies that the AOI or several AOIs in the image be created to contain primarily agricultural land-use classes. In semiarid climates, these should be irrigated agriculture to direct the statistical process to relatively potential ET rates. A target of >80% agricultural pixels in the AOI or group of AOIs is desired, with some inclusions of roadways, homesteads, small towns, nonagricultural areas, etc. permitted. The AOI should exclude significant areas of nonagricultural areas such as desert, forest, lakes, steep slopes, or cities. It is desirable to locate the AOI (s) within 20-30 km of the weather station from which the ET_r is determined for calibration. Alternatively, a series of "point" AOIs (option 1b) can be created by selecting all pixels or a subset (grid) of all pixels within 20-30 km of the weather station that have an agricultural (or irrigated in dry climates) land use (LU; for example, LU = 81, 82 for the National Land Cover Dataset) (NLCD). It is recommended that filtering be applied to ensure that selected pixels lie away from field boundaries. In the case of the hot pixel selection, a grass rangeland (in arid climates) LU, for example LU 71 in the NLCD system, can be included to ensure a sufficient number of pixels having nearly "bare soil" conditions during mid growing season.

The specification of a predominately agricultural AOI from which population statistics are evaluated is important to maintain consistency in statistics of surface temperature and vegetation indices from image scene to image scene. Otherwise, statistics will substantially change from scene to scene (i.e., from path/row to path/row) as the distribution of land-use type changes. Examples of AOIs are shown in Figure 4 for southeastern and south-central Idaho.

Homogeneity in Thermal Pixels

In Landsat images, thermal pixels have larger spatial size (120 m for Landsat 5, 60 m for Landsat 7, 100 m for Landsat 8) than the short wave pixels (30 m). In these situations, consistency of population statistics within the AOI will be improved, especially for AOI option 1b of Table 1, by filtering pixels to disclude those that lie close to field boundaries. The filtering helps ensure that thermal pixels have reduced contamination from different energy-balance conditions outside of a field. The filtering can be accomplished by calculating the standard deviation (SD) of the land use within a 7 × 7 pixel cluster (for Landsat 5) centered on the pixel being evaluated. In the case of Landsat 7, a 3 × 3 cluster can be evaluated, and for Landsat 8, a 7 × 7 cluster is appropriate. Alternatively, and as used later in the ERDAS programming example, randomly sized and shaped thermal clusters of 30 m pixels can be organized based on similarity of the digital number (DN) of the 30 m thermal band or on LST itself. These clusters of similar retrieved LST can then be evaluated in regard to uniformity of NDVI.

In assessing and filtering for uniformity in land-use type, all agricultural land-use numerics can be set to the same value (i.e., 81, 82, 71 can all be set to 81, for example) prior to calculating the SD. Pixels having a cluster SD = 0 or nearly zero likely lie away from agricultural boundaries. This analysis needs to only be done once for each year or short period of years.

To ensure that candidate pixels lie away from field boundaries within agricultural classes, one can calculate SD for clusters based on NDVI or other vegetation index and for each scene if vegetation changes over time. In current level 1, terrain corrected (L1T) Landsat products from the USGS EROS data center, thermal pixels are resampled to 30 m using cubic convolution resampling to be consistent with the 30 m reflected bands. Therefore, it is not possible to view the original domains of the thermal pixels. However, the use of the 3 × 3 or 7 × 7 clusters accounts for the potential area of overlay by the thermal pixel. A low SD for NDVI in a cluster around a pixel indicates that the original overlying thermal pixel overlaid a relatively uniform expanse of vegetation. If the SD is greater than a threshold value, it can be assumed that there is too much variation in NDVI within the polygon and those pixels are rejected. In some applications of the automated procedure, a coefficient of variation, CV, threshold of 15% has been used for NDVI of clusters, where CV is the ratio of SD to the mean (Figure 5).

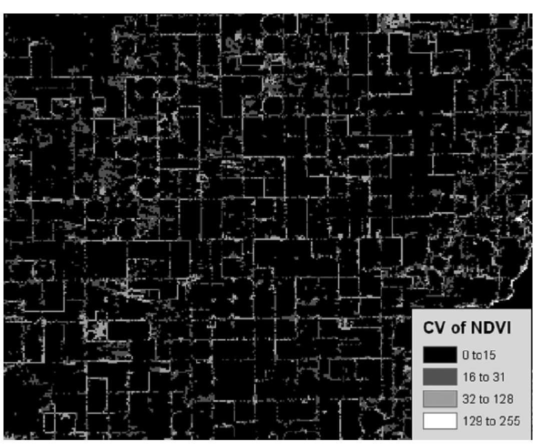

FIGURE 5. Coefficient of Variation (SD/mean) for NDVI, in Percent, Used to Filter Edges of Fields Where Contamination of Landsat Thermal Pixels Can Occur.

It is important to perform cloud and cloud-shadow identification and masking prior to the automated calibration. Clouds and shadows run substantially cooler than sunlit land surfaces and heavily biased statistics.

Cold Pixel Identification

As summarized in Table 1, the population of pixels having the top 5% of NDVI is isolated from the sample of pixels taken from relatively homogenous areas of the AOI. From this 5% subpopulation, the pixels having $T_{s_{DEM}}$ in the bottom 20% of the $T_{s_{DEM}}$ are further isolated. The isolation produces a pixel subpopulation of 1% of the original sampled and filtered (for homogeneity) population. As an example, in the top AOI example from Figure 4, pixels were sampled, followed by filtering for edge effects, using a 245 m by 330 m grid. This created over 3,100 pixel samples (Burnett, 2007), leaving about 30 pixels in the 1% isolation. The average $T_{s_{DEM}}$ is calculated from that population and cold pixel candidates are selected among the pixels that have a value for $T_{s_{DEM}}$ that is within ±0.2 K of the statistically identified average $T_{s_{DEM}}$ and have an albedo that is within ±0.02 of the albedo threshold calculated using Equation (7):

$$\alpha = 0.001343\,\beta + 0.3281\,\exp(-0.0188\,\beta) \qquad (7)$$

where α is the albedo threshold, and β is the sun angle (°) above the horizon. The form of Equation (7) was developed by Dong *et al.* (1992) for clipped grass and was parameterized in this study using albedo sampled from METRIC over a range of solar angles and agricultural crop types in southern Idaho (Ricardo Trezza and Richard Allen, unpublished data). The purpose of filtering cold pixel candidates using albedo from Equation (7) is to limit the range of albedo used in the selection. This is done because of the relatively strong impact of albedo on available energy at the surface and therefore on the equilibrium surface temperature for transpiring vegetation. Generally, a recommended range for albedo for the cold pixel candidate is 0.18-0.25, but these values increase during non-summer months with lower β at image time, which is reflected in Equation (7). The albedo of the cold pixel candidate should remain in this range, since it is more likely to be consistent with the ET rate represented by the alfalfa reference ET estimate (which assumes a constant albedo of

0.23). The $\pm$0.02 range for albedo from the estimate by Equation (7) can be relaxed or shifted as needed to fit the particular vegetation in the image.

Hot Pixel Identification

As summarized in Table 1, for the hot pixel selection, the population of pixels having the lowest 10% of all NDVI is isolated from the subsample of homogenous pixels from the filtered AOI. From this 10% subpopulation, pixels having $T_{s_{DEM}}$ in the top 20% are isolated. This isolation results in a pixel population of 2% of the original sampled and filtered (for homogeneity) population. This 2% will have nearly the lowest NDVI and the hottest $T_{s_{DEM}}$. From this 2% population, the average $T_{s_{DEM}}$ is calculated and represents the target $T_{s_{DEM}}$ for the hot pixel to search for. Earlier comments regarding recommended albedo for the hot pixel and elimination of any pixel candidates impacted by organic residue apply here.

Comparisons with manually selected $T_{s_{DEM}}$ in southern Idaho climates indicate that this average $T_{s_{DEM}}$ does require an adjustment, however, during periods of very low rainfall and therefore extreme dryness of nonirrigated soils that may lie in the AOI, where those soils may be substantially drier than bare, agricultural soils to which the ET_rF_{hot} will be assigned. The differences in dryness translate into differences in the mean $T_{s_{DEM}}$ computed for the isolated subpopulation. During periods of some precipitation, the statistical process is relatively accurate, without adjustment.

The mean $T_{s_{DEM}}$ of the isolated hot pixel candidates is modified by subtracting a temperature offset that is computed as a function of the ratio of precipitation to reference ET over the previous 60 days, as developed by Allen and Burnett (2006, internal report):

$$T_{\text{fac}} = 2.6 - 13\frac{\text{precipitation}_{60}}{\text{ET}_{r_{60}}} \tag{8}$$

where T_{fac} is the adjustment offset to be subtracted from the mean $T_{s_{DEM}}$ selected statistically for the hot pixel, precipitation$_{60}$ is the precipitation over the previous 60 days, mm, and $\text{ET}_{r_{60}}$ is the alfalfa reference ET over the previous 60 days, mm. The offset is limited to ≥ 0. When the ratio precipitation/reference ET is greater than about 0.2, no adjustment is needed. Equation (8) was derived primarily for application in a semi-arid or arid climates in temperate latitudes having low amounts of precipitation during the growing season. It may not be valid, or necessary, for application during winter periods or in nontemperature zones.

Figure 6 shows the data used to determine the shape and coefficients for Equation (8), based on com-

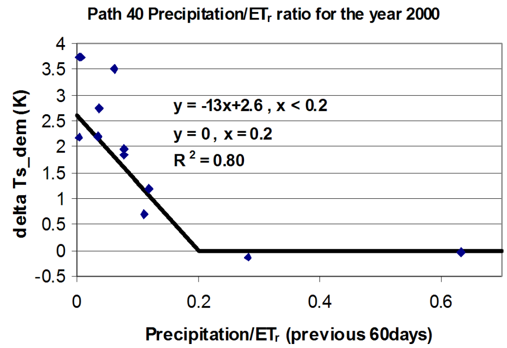

FIGURE 6. Difference Between the $T_{s_{DEM}}$ of the Hot Pixel Candidates Selected by the Automated Procedure of Table 1 and That of the Hot Pixel That Was Manually Determined for 12 Landsat Image Dates During 2000 in Southern Idaho (Landsat path 40, row 30).

parisons of manual *vs.* automated selection of hot pixels from 12 image dates for path 40 row 30 of Landsat during year 2000. The "delta T_s_dem" in the figure is the difference between the $T_{s_{DEM}}$ of the hot pixel candidates selected by the automated procedure and that of the hot pixel that was manually determined.

Hot pixel candidates are selected among the pixels that meet the requirements of lying in the interiors of homogenous vegetation having close to bare conditions that is, having no mulch or vegetation residue cover, and having a value for $T_{s_{DEM}}$ that is within $\pm$0.2 K of the statistically identified average $T_{s_{DEM}} - T_{\text{fac}}$. Vegetation residue can be identified, and pixels eliminated, using the mid-infrared bands 5 and 7 of Landsat, following, for example, the approach of Aggarwala *et al.* (1991) or by eliminating pixels having low NDVI and high albedo exceeding that common to dry soil, for example, 0.30-35. Otherwise, the selection of the hot pixel is not substantially influenced by the albedo of the pixel because, generally, as albedo for a dry pixel increases, the $T_{s_{DEM}}$, H, and dT decrease, proportionately, due to less net radiant energy at the surface. Low NDVI fields having organic cover need to be eliminated from the calibration selection because of the insulative effect of the residue on reducing soil heat flux, which may not be captured accurately by the G estimate and by increased uncertainty in any residual evaporation following a wetting event.

Setup of the Automated Anchor Pixel Selection Procedure Using ERDAS Modelmaker

Table 2 summarizes a coding strategy to implement the automated pixel selection strategy of Table 1 into an ERDAS Imagine Modelmaker process. Other processes can be employed, for example

TABLE 2. Auto Calibration Pixel Selection Procedures as Implemented into ERDAS Imagine Modelmaker Code.

Model	Input	Output	Description
1	Band 6 from Landsat	Four identical raster maps: Mean, SD, CV, and CV15	In model 1, thermal pixels of an image are grouped into polygons having very similar digital numbers or similar LST to create polygons of essentially uniform surface temperature. CV15 is the CV times 0.15
2	Maps of NDVI, Mean, and SD	Attributes to Mean, SD, CV, and CV15	In model 2, the mean, standard deviation, and coefficient of variation of NDVI for each common thermal polygon is calculated and attributed to the rasters created in model 1. A binary map indicating whether the CV in NDVI within each polygon is above or below a filtering threshold of 15% is created as the end product
3	Maps of CV15, NDVI, T_s, and albedo	Map of cold pixel candidates	Candidates for cold anchor pixels are calculated in model 3. The albedo threshold value is calculated using Equation (7)
4	Maps of CV15, NDVI, and T_s	Map of hot pixel candidates	Candidates for hot anchor pixels are calculated in model 4. The T_{fac} adjustment calculated using Equation (8) is included in the model
5	Maps of cold and hot pixel candidates	Maps of cold and hot pixel candidates (in color)	A colorization routine is applied to indicate recommended anchor pixel candidates, for example using green to indicate more preferred pixels and red less preferred pixels within the final subset of anchor pixels candidates. The user can review these candidates to select a pixel located in the interior of a field with relatively homogeneous vegetation. Alternatively, for complete automation, the candidate pixel lying closest to the weather station used to calibrate METRIC can be selected

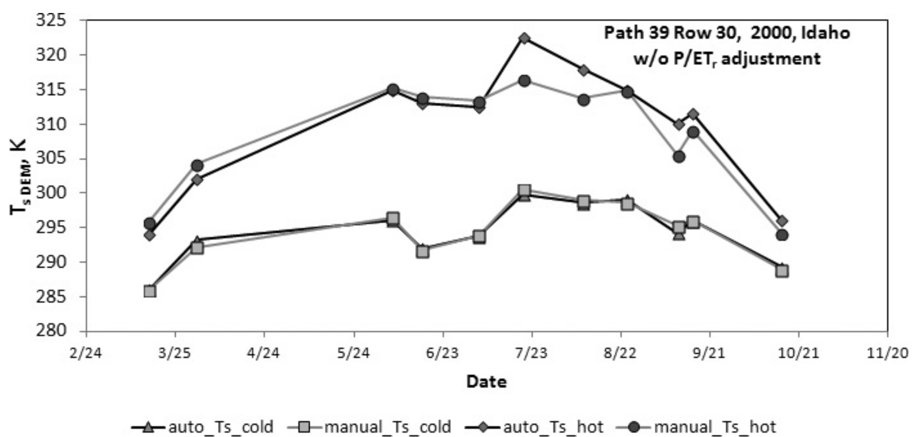

FIGURE 7. $T_{s_{DEM}}$ for Hot and Cold Calibration Pixels via the Automated Pixel Selection Method and Manual Selection for Image Dates During 2000 in Landsat WRS Path 39, Row 30 of Southeastern Idaho.

Python, as done by Morton *et al.* (2013). Following the automated processing, the user should review the potential calibration pixel candidates carefully. The visual check is needed to preclude selecting pixels located in nonagricultural inclusions such as feedlots, construction zones, and orchards.

COMPARISONS IN STUDY AREAS

Figures 7-10 illustrate the performance of the automated pixel selection method in comparison to manual selection of calibration pixels for applications along Landsat WRS paths 39 and 40 of southern Idaho, path 41 of western Montana, and further east along paths 35 and 36. Each figure compares $T_{s_{DEM}}$ for hot and cold pixels for the series of Landsat images processed. Temperatures of calibration pixels varied from image date to image date due to random change in weather characteristics and in some cases due to background evaporation from precipitation. Similar comparisons of the automated selection method with manually determined selections have been made in Nevada by Morton *et al.* (2013) with relatively good and consistent results.

Figures 7 and 8a show comparisons for southern Idaho where the statistics were evolved without application of the adjustment of Equation (8) for regional dryness. In those figures, agreement between $T_{s_{DEM}}$ for the automated (statistical) and manually selected calibration pixels was very good for cold pixel endpoints over all image dates and was good for hot pixel endpoint selections for periods outside of the late June through August period when

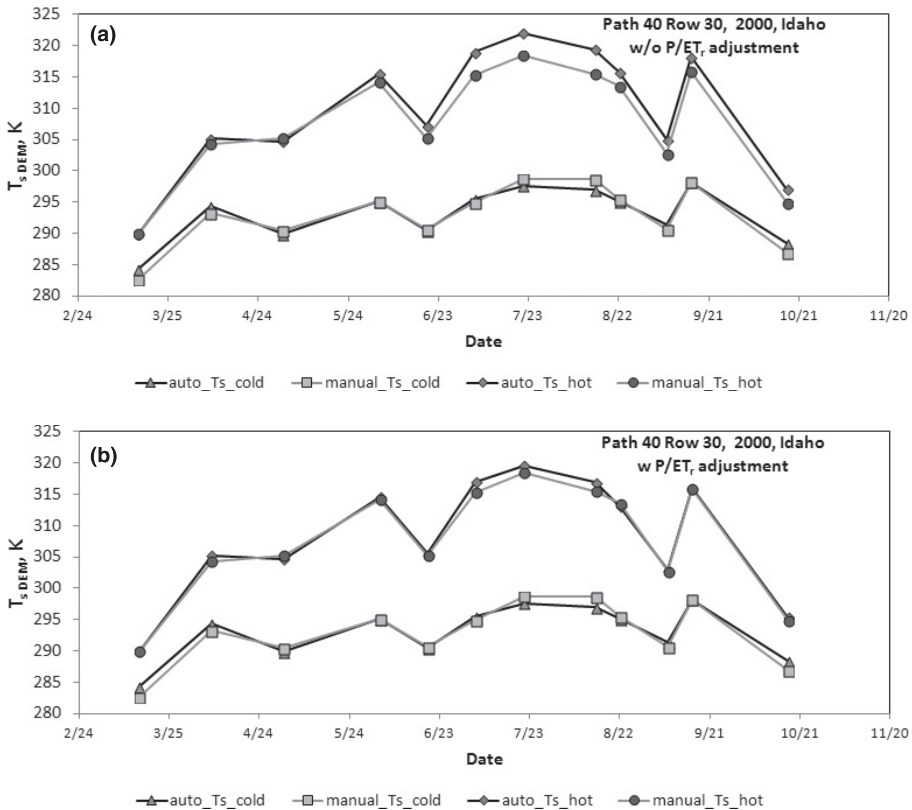

FIGURE 8. (a) $T_{s_{DEM}}$ for Hot and Cold Calibration Pixels via the Automated Pixel Selection Method and Manual Selection for Image Dates During 2000 in Landsat WRS Path 40, Row 30 of Southern Idaho Prior to Adjustment of the $T_{s_{DEM}}$ for the Hot Pixel Using Equation (8). (b) $T_{s_{DEM}}$ for hot and cold calibration pixels via the automated pixel selection method and manual selection for image dates during 2000 in Landsat WRS path 40, row 30 of southern Idaho following adjustment of the $T_{s_{DEM}}$ for the hot pixel using Equation (8).

conditions in Idaho are hot and dry. Figure 8b compares the same images and calibration endpoints as Figure 8a except that Equation (8) was applied to the $T_{s_{DEM}}$ for the hot pixels isolated by the automated procedure. The adjustment lowered the recommended value for $T_{s_{DEM}}$ for image dates during the late June-August period by about 2.5 K and improved agreement with manually derived temperatures. The absolute average deviation for cold pixels was 0.4 K for path 39 and 0.8 K for path 40 and for hot pixels was about 2.2 K for both paths before application of the Equation (8) adjustment. Following application of the adjustment from Equation (8), the absolute average deviation for the hot pixels was 0.6 K. Burnett (2007) analyzed recalculations with METRIC using these errors for specified $T_{s_{DEM}}$ for cold and hot endpoint pixels and concluded that a deviation in $T_{s_{DEM}}$ of 0.8 K would produce an error of less than 2% of average ET flux estimated for fully vegetated conditions. Burnett (2007) concluded that a 2.0 K deviation in $T_{s_{DEM}}$ for the hot pixel candidate would produce an average error in sensible heat flux, H, of around 33 W/m^2, equivalent to about 7% of reference ET_r.

The average 0.6 K average error for hot pixel selection would cause about 2% error in ET_rF. This error is considered to be small relative to error common to ET estimation using crop coefficient-reference ET approaches and common to ET measurement, for example via lysimeter and eddy covariance, that tends to average about 10% in practice, if done well (Allen *et al.*, 2011b). These analyses suggest about 3-4% error in ET per 1 K error in hot or cold pixel T_s assignment during mid growing season conditions. Similar estimates for error in ET that stem from any error in the value specified for $T_{s_{DEM}}$ during hot and cold pixel selection can be made by dividing the 0.6-0.8 K average error that is expected to occur between values selected during the automated selection and the manual selection by the range between the $T_{s_{DEM}}$ value for the hot and cold calibration pixels. These latter values are displayed in Figures 6-9. During midseason, that range averages about 20 K so that expected error in H would be about 3-4% which, for H~0 for fully vegetated conditions and H~$R_n - G$ for dry conditions, would suggest error in relative ET (ET_rF) of 3-4%.

Errors in the hot pixel selection affect the final ET determination primarily toward the lower end of the ET spectrum and have little impact on relatively wet pixels, and vice versa. Stated differently, the two extreme pixels and conditions tend to impact accuracy

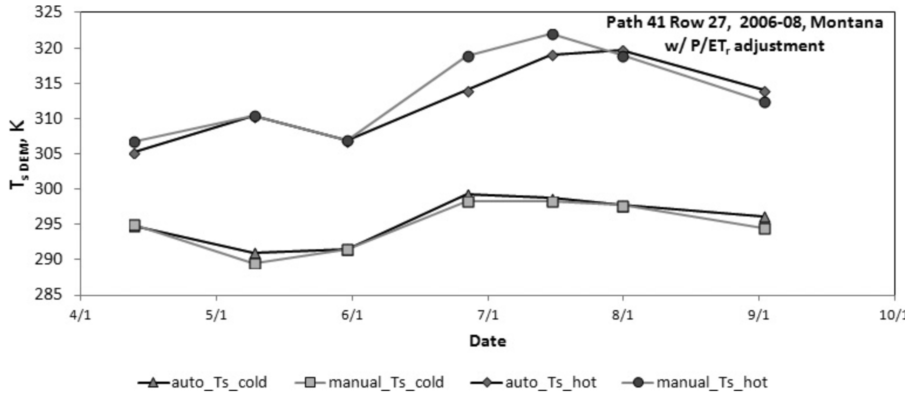

FIGURE 9. $T_{s_{DEM}}$ for Hot and Cold Calibration Pixels via the Automated Pixel Selection Method and Manual Selection for Seven Landsat 5 Image Dates During 2006-2008 in Landsat WRS Path 41, Row 27, Located in Northwestern Montana.

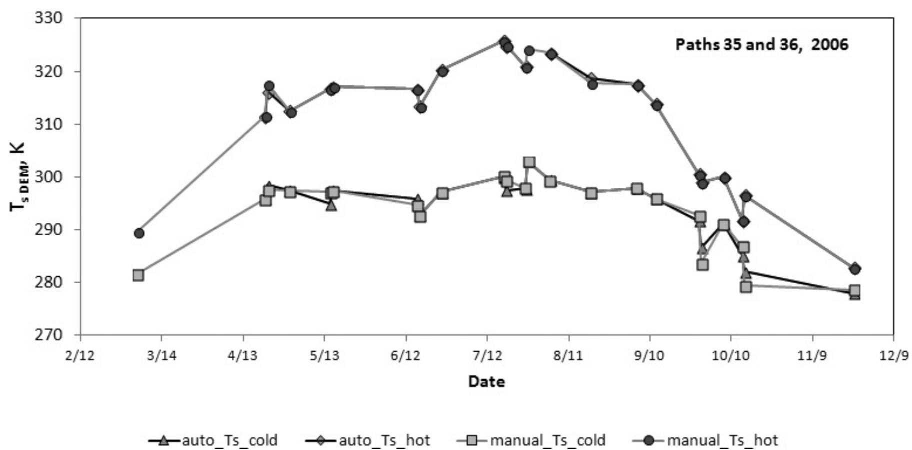

FIGURE 10. $T_{s_{DEM}}$ for Hot and Cold Calibration Pixels via the Automated Pixel Selection Method and Manual Selection for 23 Landsat 5 and 7 Image Dates During 2006 Along Landsat WRS Paths 35 and 36.

of the ET estimates in the domain of those conditions, with diminishing impacts toward the other end of the ET spectrum.

Figure 9 shows results of automated selection for path 41 of western Montana (Kjaersgaard and Allen, 2008). The area analyzed is in the Mission Valley containing a 20 km wide and 35 km long area dominated by irrigated agriculture and surrounded by mountains. The absolute average deviation between $T_{s_{DEM}}$ for cold calibration pixels was 0.7 K, which is similar to the deviation found for southern Idaho. The absolute average deviation between $T_{s_{DEM}}$ for hot calibration pixels is 1.7 K, which is estimated to translate to about 5-10% error in ET_rF. The good agreement for the first image date, April 13, is promising as there were few fields having near full vegetation cover by that early date.

Figure 10 shows results of automated selection for areas along Landsat WRS paths 35 and 36. The AOI used for this application of the automated process followed option 2 of Table 1 where a series of point AOIs was created from irrigated agricultural land-use classes. The absolute average deviation between $T_{s_{DEM}}$ for cold calibration pixels was 0.7 K, which is similar to the deviation found for southern Idaho and absolute average deviation for hot calibration pixels was only 0.1 K. These errors are expected to translate to less than 3% error in ET_rF.

CONCLUSIONS

Consistency in distribution of vegetation indices such as NDVI and surface $T_{s_{DEM}}$ within predominately agricultural areas provides a useful means to isolate relatively dependable and accurate calibration endpoints for a *CIMEC* calibration process used in the METRIC ET model, when applied to 30 m resolution Landsat images. The relatively accurate identification of hot and cold calibration endpoints for $T_{s_{DEM}}$ for environmental contexts and AOI forms for paths 35, 36, and 41 that differed from those of the development areas in Idaho suggest that the isolation target statistics of 5 and 10% NDVI and 20% T_s for the cold and hot conditions may work for relatively general applications. Expected error in the automated calibration procedures is expected to be around 10% relative to manual calibration by experienced, cognizant users when applied to climates similar to those of the Rocky Mountain region (semiarid climates with dry summers).

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